ISTANBUL AREL UNIVERSITY
FACULTY OF ENGINEERING

LEEN364-DEEP LEARNING AND CLASSIFICATION TECHNIQUES

PROJECT TEMPLATE

Project Title: Smart Drug Recognition

Group Number: 8
Group Members:

- Youssef Abdalla - 220303958
  - Dataset preparation, data pipeline design, preprocessing script (prepare_dataset.py), project integration) .
- Ahmed Ali - 20303975
  - *Model selection, MobileNetV2 architecture design, training strategy, fine-tuning.*
- Ali Al-Mustafa - 220303971
  - *Training analysis, evaluation metrics, confusion matrix, performance interpretation.*
- Ahmed Bahgat - 220303923
  - *User interface design, prediction visualization, project presentation and documentation.*

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1. INTRODUCTION

- **Objective**
  - The objective of this project is to develop a deep learning–based image classification system capable of distinguishing between **original** and **counterfeit medicine packages** using computer vision techniques.
  - The system takes an image of a drug package as input and outputs a classification result along with a confidence score, helping to identify potentially counterfeit medicines.
- **Motivation**
  - Counterfeit medicines represent a serious global health problem. Fake drugs may contain incorrect ingredients, wrong dosages, or harmful substances, posing significant risks to patients.
  - Since counterfeit packages are often visually similar to original ones, manual inspection is unreliable. Therefore, an automated image-based solution using deep learning is both necessary and impactful

2. LITERATURE REVIEW

- Traditional image classification methods such as **k-Nearest Neighbors (k-NN)**, **Logistic Regression**, and handcrafted feature extraction techniques have been used in earlier image recognition tasks. However, these methods struggle with complex visual patterns and subtle differences.
  Example:
- *Recent studies show that **Convolutional Neural Networks (CNNs)** significantly outperform traditional methods in image-based classification tasks due to their ability to learn hierarchical visual features.*
- Additionally, **Transfer Learning** models such as MobileNetV2, ResNet, and VGG have proven effective when working with limited datasets by leveraging pretrained knowledge from large-scale datasets like ImageNet.

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3. DATASET DESCRIPTION

- **Dataset Name**: Fake vs Real Medicine Dataset.
- **Source**: Kaggle Dataset: https://www.kaggle.com/datasets/surajkumarjha1/fake-vs-real-medicine-datasets-images
- Dataset Overview:
  - *The dataset consists of approximately **661 images**, including:*
    - ***421 Original medicine packages***
    - ***240 Counterfeit medicine packages***
  - *Images include medicine boxes, strips, packs, and labels captured under different conditions.*
- **Features**:
  - Input: RGB images resized to **224 × 224**
  - Output labels:
    - original
    - counterfeit
- **Preprocessing**:
- The following preprocessing steps were applied:
  - Image resizing to 224×224 pixels
  - Normalization using MobileNetV2 preprocessing
  - Automated dataset splitting using a custom script
  - No image appears in more than one split (no data leakage)
- Dataset Split (70 / 15 / 15)
  - **Training:** 462 images
  - **Validation:** 99 images
  - **Test:** 100 images

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4. METHODOLOGY

- Describe your approach step by step.
    - **Model Selection**:
      We used **MobileNetV2** with Transfer Learning due to its efficiency, lightweight architecture, and strong performance on image classification tasks with limited data.
    - **Frameworks and Tools**:
        - TensorFlow
        - NumPy
        - Matplotlib
        - Scikit-learn
    - **Training Strategy**:
        - The training was performed in two phases:
            - Phase 1 – Feature Extraction
                - MobileNetV2 base frozen
                - Only the classification head was trained
            - Phase 2 – Fine–Tuning
                - Last 30 layers of MobileNetV2 unfrozen
                - Very small learning rate to adapt features safely

        - Hyperparameters
            - Batch size: 32
            - Optimizer: Adam
            - Learning rate
                - Phase 1: 0.001
                - Phase 2: 0.00001
            - Epochs: 30

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5. CODE IMPLEMENTATION

- The project code is structured into modular scripts:
- Prepare_dataset.py
    - Handles merging, shuffling, splitting, renaming, and summarizing the dataset.
- Train_model.py
    - Builds the MobileNetV2 model, trains it in two phases, and saves the trained model.
- Evaluate_model.py
    - Evaluates the model on the test set and generates accuracy, confusion matrix, and classification reports.
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6. RESULTS and ANALYSIS

- Test Perfrmance
  - Test Accuracy: 97.0%
  - Test Loss: 0.083
- Classification Metrics
  - Precision: 0.97
  - Recal: 0.96
  - F1-Score**:** 0.97

Confusion Matrix

Actually / Predicted	**Counterfeit**	**Original**
Counterfeit	34	2
Original	1	63

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7. DISCUSSION and CONCLUSION

- **Performance Evaluation**:
  - The model achieved high accuracy and stable validation performance, demonstrating effective generalization. Fine-tuning significantly improved the model's ability to capture subtle packaging.
- **Challenges**
  - Limited dataset size
  - High visual similarity between classes
  - Training instability during fine-tuning
- **Solutions**
  - Transfer Learning
  - Controlled fine-tuning
  - Early stopping and learning rate scheduling
- **Conclusion**
  - This project successfully demonstrates how deep learning and computer vision can be applied to detect counterfeit medicines. The system achieved high accuracy and can be extended to real-world applications.
- **Future work**
  - Larger and more diverse datasets
  - Multi-class classification
  - Real-time mobile deployment
  - Integration with barcode or text recognition

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8. REFERENCES

- Kaggle Fake vs Real Medicine Dataset

- TensorFlow Documentation
- MobileNetV2 Paper
- Scikit-learn Documentation

8. PROJECT LINK:

- Link: https://github.com/youssefhany29/Smart-Drug-Recognition-System.git